CAN MACHINE-LEARNING AND DEEP-LEARNING MODELS FORECAST DAILY EQUITY VOLATILITY BETTER THAN CLASSICAL BASELINES?

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ABSTRACT

This report presents a self-guided term project on one-day-ahead volatility forecasting for equity indices. Volatility drives option pricing, value-at-risk and position sizing, and, unlike the direction of returns, is strongly persistent and therefore forecastable. The objective was to test whether machine-learning (gradient-boosted trees) and deep-learning (LSTM) models can outperform well-implemented classical baselines. Eight models (naive, 21-day rolling variance, RiskMetrics EWMA, GARCH(1,1), GJR-GARCH, EGARCH, a walk-forward XGBoost and a regularized LSTM) were evaluated on 18.5 years of S&P 500 daily returns (2008–2026) under a strict walk-forward protocol with no look-ahead bias, using the noise-robust QLIKE loss and the Diebold–Mariano test. A monthly re-estimated EGARCH(1,1) achieved the best out-of-sample QLIKE (1.504), statistically tied with GJR-GARCH ($p = 0.47$); both leverage models significantly beat plain GARCH(1,1) ($p < 0.001$) on a test window containing the 2022 bear market and the April-2025 tariff shock. The LSTM ranked third, ahead of plain GARCH and statistically indistinguishable from the winner ($p = 0.19$), and the learned models attained the lowest RMSE (LSTM) and MAE (XGBoost). Even so, no learned model outranked the leverage GARCH family on the risk-relevant criterion: the classic finding that GARCH-type models are extremely hard to beat on daily data survives, in leverage-aware form.

Index Terms: Volatility forecasting, GARCH, QLIKE, XGBoost, LSTM, walk-forward evaluation

1. INTRODUCTION

Volatility, the conditional standard deviation of an asset's returns, is the central quantity of financial risk management: it prices options, sets value-at-risk limits and margin requirements, and determines position sizes. Forecasting it accurately, even one day ahead, has direct practical value for any trading desk or risk function.

The problem is scientifically interesting because of a sharp asymmetry in predictability. The *direction* of tomorrow's return is essentially unpredictable, but its *magnitude* is not: calm days cluster with calm days and turbulent days with turbulent days. Classical econometrics has exploited this persistence for four decades, most famously through the GARCH family [1]. In recent years, machine-learning and deep-learning models have been proposed as replacements, with mixed and sometimes contested results [7].

This project puts that proposition to a controlled test. Classical baselines and learned models were implemented within a single evaluation harness on 18.5 years of S&P 500 daily data, with identical information sets, a shared out-of-sample window, a proxy-noise-robust loss and a formal significance test, so that any claimed improvement must survive statistical scrutiny rather than visual inspection.

Section 2 formulates the problem and lists the objectives. Section 3 details the methodology: the baselines, the learned models and the walk-forward protocol. Section 4 presents the experimental setup and results. Section 5 discusses and interprets the findings, Section 6 concludes, and Section 7 links the project artifacts.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Let $r_t = \ln(P_t/P_{t-1})$ be the daily log return of an asset, modelled as $r_t = \sigma_t z_t$ with z_t zero-mean, unit-variance noise. The task is **one-day-ahead variance forecasting**: at the close of day $t-1$, output a forecast $\hat{\sigma}_t^2$ of day t's conditional variance using information up to day $t-1$ only.

The fundamental difficulty is that true volatility is *latent*: it is never observed on any day. Forecasts must be scored against a proxy; this project uses the squared return r_t^2, which is conditionally unbiased ($\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$) but extremely noisy. The noise dictates two design constraints adopted throughout: (i) the headline evaluation loss must be robust to proxy noise, which motivates QLIKE [9]; and (ii) learned models cannot regress directly on r_t^2 without collapsing, which motivates a smoothed learning target (Section 3.2).

Figure 1 motivates the entire study: over the test window, the autocorrelation function of raw returns is flat (no linear predictability in the mean), while the ACF of absolute returns decays slowly and remains significant for dozens of lags. That persistence in magnitude is the only signal any model below exploits.

The scope is deliberately constrained: a single liquid index (S&P 500), daily frequency, price-derived features only (no exogenous data), and a one-day horizon.

2.2. Objectives

1. To implement honest classical baselines, namely naive, rolling, EWMA and the GARCH family (GARCH, GJR-GARCH, EGARCH) with periodic re-estimation, as the standard to beat.

2. To design a leak-proof walk-forward evaluation harness

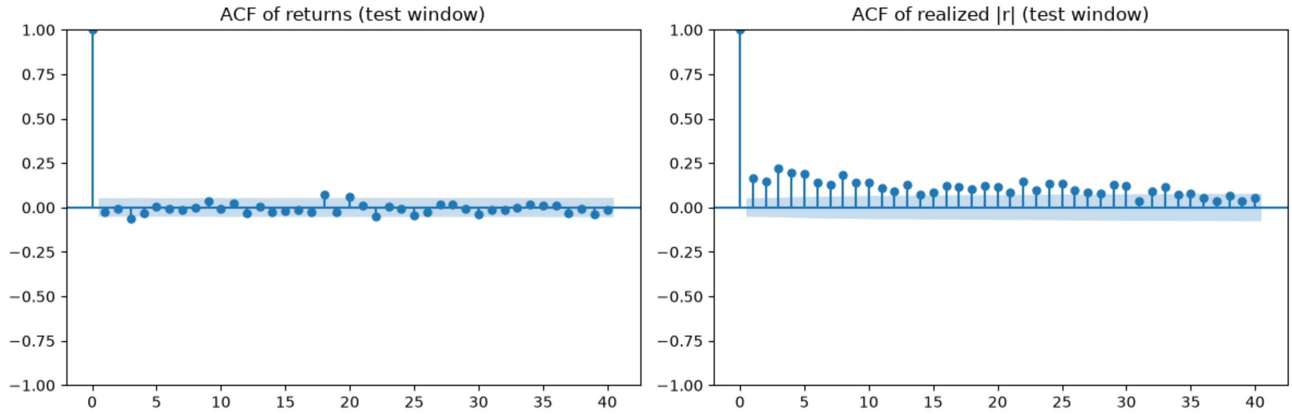

Fig. 1. Correlograms over the test window: ACF of returns (left) and of absolute returns (right), 40 lags with 95% confidence bands. Returns are serially uncorrelated; absolute returns show slowly decaying, strongly significant autocorrelation, i.e. volatility clustering, the signal all models exploit.

with RMSE, MAE, QLIKE and the Diebold–Mariano test.

3. To design and implement a walk-forward XGBoost forecaster with justified features and target.
4. To design, implement and regularize an LSTM forecaster with explicit anti-leakage guards.
5. To evaluate all models on a common out-of-sample window and determine whether observed differences are statistically significant.
6. To deliver a reproducible, modular codebase.

3. METHODOLOGY / APPROACH

3.1. Overall Workflow

The pipeline runs end to end from raw prices to a significance-tested leaderboard: (1) daily closes are downloaded and converted to log returns; (2) the final 30% of days is frozen once as the out-of-sample test window, shared by every model; (3) each model produces one-day-ahead variance forecasts for the test window, using information up to the previous day only (classical models via recursions re-estimated periodically, learned models via walk-forward re-training); (4) forecasts are scored against the squared-return proxy with RMSE, MAE and QLIKE; (5) the best model is compared with the best remaining baseline by the Diebold–Mariano test. The five expensive model fits (three GARCH variants, XGBoost, LSTM) are mutually independent and run in parallel worker processes.

3.2. Technical Details

Classical baselines. The naive forecast is $\hat{\sigma}_t^2 = r_{t-1}^2$; the rolling baseline is the variance of the previous 21 returns; EWMA is the RiskMetrics recursion $\hat{\sigma}_t^2 = \lambda\hat{\sigma}_{t-1}^2 + (1-\lambda)r_{t-1}^2$ with $\lambda = 0.94$ [5]. The GARCH family models conditional variance as

$$\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2, \tag{1}$$

estimated by maximum likelihood (zero mean, normal innovations). GJR-GARCH [3] adds the threshold term $\gamma\, r_{t-1}^2\, \mathbf{1}[r_{t-1} < 0]$ to capture the leverage effect; EGARCH [2] models $\ln \sigma_t^2$ with a signed standardized-residual term, guaranteeing positivity by construction. All three are re-estimated every 22 trading days on an expanding window; between refits the fitted recursion is rolled forward daily with observed returns. The hand-rolled recursions were verified against the reference implementation's internal source so that daily updates match the fitted parameters exactly.

Features and target (shared by both learned models). Features at row t use data through $t-1$ only: lagged returns and squared returns (lags 1–10), realized variance over the past 5 and 21 days (spanning the multi-horizon structure of the HAR model [8]), and the 5-day mean absolute return. Signed return lags let the models discover the leverage effect on their own. The learning target is

$$y_t = \ln\left(\tfrac{1}{5}\textstyle\sum_{j=0}^{4} r_{t+j}^2 + \varepsilon\right), \tag{2}$$

the log of forward 5-day average squared returns: averaging suppresses proxy noise and the log makes the target approximately Gaussian. Because y_t is only observed at $t+4$, every training set excludes a 5-day buffer of the most recent rows.

XGBoost. A gradient-boosted ensemble (400 trees, depth 3, learning rate 0.03, row/column subsampling 0.8, a deliberately conservative low-variance configuration) is re-trained from scratch every 22 days on all fully realized targets, mirroring the GARCH refit cadence so the comparison is fair. Warm-starting refits from the previous booster was evaluated and rejected: it changes ensemble composition for negligible savings, sacrificing reproducibility.

LSTM. A single-layer LSTM (32 hidden units, linear head) reads the previous 20 days of standardized features and predicts y_t. Anti-leakage guards are explicit: sequences are labelled by their final row; standardization uses training-window statistics only; the early-stopping validation slice is carved from the *end of the training window*, never from test. Training uses Adam on MSE in log-variance space, mini-batches of 64, up to 200 epochs with patience 15 and best-checkpoint restoration, and a fixed seed. The architecture is deliberately small: roughly 2,800 noisy training sequences cannot support a deep stack.

Table 1. Out-of-sample leaderboard, test window 2020-12-07 to 2026-06-29 (1,395 days), sorted by QLIKE (lower is better). RMSE and MAE on the volatility scale; QLIKE on the variance scale. Best value per column in bold.

Model	N	RMSE	MAE	QLIKE
EGARCH(1,1)	1395	0.00728	0.00556	**1.5042**
GJR-GARCH(1,1)	1395	0.00731	0.00554	1.5091
LSTM	1387	**0.00686**	0.00499	1.5287
GARCH(1,1)	1395	0.00751	0.00576	1.5569
EWMA(0.94)	1395	0.00750	0.00573	1.5814
XGBoost	1395	0.00691	**0.00496**	1.5911
Rolling(21)	1395	0.00762	0.00571	1.6344
Naive	1395	0.00951	0.00672	5440.50

Evaluation losses. RMSE and MAE are computed on the volatility scale ($\sqrt{\text{variance}}$) for interpretability. The headline criterion is

$$\text{QLIKE} = \frac{1}{T}\sum_t \left(\frac{p_t}{v_t} - \ln\frac{p_t}{v_t} - 1\right), \tag{3}$$

with proxy $p_t = r_t^2$ and forecast $v_t = \hat{\sigma}_t^2$: QLIKE rankings are robust to noise in an unbiased proxy [9], and the loss penalizes under-prediction of variance more heavily than over-prediction, the asymmetry a risk manager wants. Loss differences are tested with the Diebold–Mariano statistic [4] with the Harvey–Leybourne–Newbold small-sample correction [6] on per-day QLIKE differentials.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

Data. S&P 500 index (`^GSPC`) daily closes, 2008-01-01 to 2026-06-30, from Yahoo Finance (auto-adjusted), yielding 4,650 daily log returns from 2008-01-03 to 2026-06-29. The sample spans the 2008 financial crisis, the 2018 "volmageddon", the COVID-2020 crash, the 2022 bear market, the April-2025 tariff shock and several calm regimes.

Split. The final 30% of observations forms the out-of-sample test window: 3,255 training days (2008-01-03 to 2020-12-04) and 1,395 test days (2020-12-07 to 2026-06-29). The COVID crash falls in the training window; the test window contains the 2022 bear market and the April-2025 tariff shock, so models are stress-tested out of sample on two sharp drawdowns.

Environment. Windows 11, CPU only; Python 3.13 with `numpy` 2.4.4, `pandas` 3.0.2, `arch` 8.0.0, `xgboost` 3.3.0, `torch` 2.12.1, `statsmodels` 0.14.6. All seeds fixed (42); one shared split; the full pipeline executes in about one minute with the five heavy fits in parallel processes.

4.2. Results

Table 1 reports the leaderboard over the 1,395-day test window, sorted by QLIKE. The LSTM covers 1,387 of 1,395 days because its first prediction requires a full 20-day feature sequence. The naive baseline's QLIKE of 5.44×10^3 reflects a property of the loss, which diverges whenever a near-zero variance forecast meets a non-zero realization.

Figure 2 overlays every model's one-day-ahead volatility forecast on realized $|r_t|$ across the test window; Figure 3 isolates the leaderboard winner against realized volatility. Diebold–Mariano tests on per-day QLIKE give three results. Between the two best models, EGARCH(1,1) versus GJR-GARCH(1,1), the statistic is -0.725 with $p = 0.468$: the two leverage models are statistically tied. Between EGARCH(1,1) and plain GARCH(1,1) the statistic is -4.211 with $p < 0.0001$: the leverage models' advantage over the symmetric recursion is highly significant. Between EGARCH(1,1) and the LSTM the statistic is -1.302 with $p = 0.193$: the LSTM is statistically indistinguishable from the winner. The LSTM early-stopped at epoch 37 of a 200-epoch budget with best validation MSE (log-variance) of 0.898.

5. DISCUSSION

Why the leverage models now lead. The test window is dominated by two sharp drawdowns, the 2022 bear market and the April-2025 tariff shock, and the leverage effect is precisely the mechanism that turns a large negative return into higher next-day variance. GJR's threshold term and EGARCH's signed-residual term encode it directly, and on this window their advantage over the symmetric GARCH(1,1) recursion (1) is highly significant ($p < 0.0001$). This is a notable shift from a shorter sample ending in 2024, where the same comparison was insignificant: the leverage effect earns its keep exactly when the out-of-sample period contains asymmetric crashes.

Why the learned models still do not win. Three structural reasons persist. First, maximum likelihood estimates the GARCH family against the correct variance objective, whereas the learned models train on the smoothed surrogate (2), which is necessary to avoid proxy-noise collapse but biases them toward smoothness. Second, roughly 3,200 noisy training observations is a small-data regime that favours 3–4 parameter recursions over 32-unit networks; reported deep-learning wins in this literature typically involve intraday realized-variance data, large cross-sections of assets or exogenous information. Third, QLIKE's asymmetry (3) rewards fast upward reactions: GARCH-type recursions respond to a single large return the next day, while tree ensembles and MSE-trained sequence models arrive late to eruptions and are charged heavily for it. That said, the case against the learned models is weaker on this longer sample: the LSTM, whose signed return lags let it learn the leverage pattern from data, ranks third, ahead of plain GARCH, and is statistically indistinguishable from the winner ($p = 0.19$).

Trade-offs observed. The learned models' best-in-class point accuracy (LSTM the lowest RMSE, XGBoost the lowest MAE) alongside a QLIKE deficit (Table 1) is the clearest trade-off: the smoothed target makes them accurate on average but slow at spikes, and the two metric families price those errors differently. Warm-starting XGBoost refits traded reproducibility for negligible speed and was rejected.

Limitations. (i) Single asset and single test window: conclusions concern the S&P 500 over 2020–2026, not ML-for-volatility in general. (ii) The squared-return proxy is the noisiest admissible choice; intraday realized variance would sharpen both training and evaluation. (iii) Hyperparameter search was deliberately modest, so the learned models'

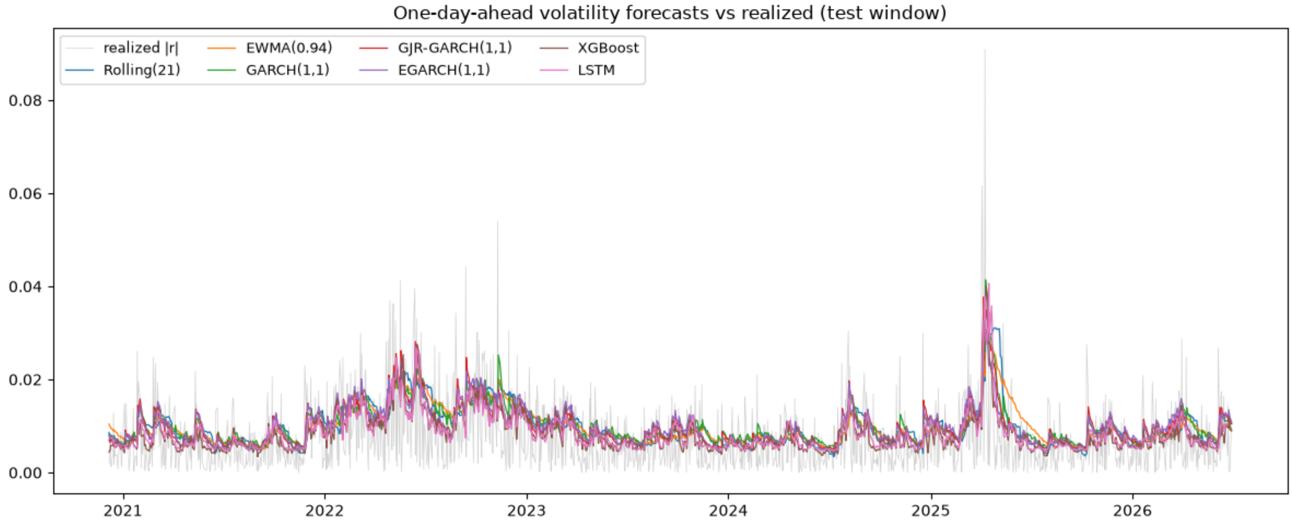

Fig. 2. One-day-ahead volatility forecasts of all models (naive omitted for legibility) against realized $|r_t|$ (grey), test window 2020-12-07 to 2026-06-29. All models track the same volatility clusters; they differ in smoothness and in responsiveness at eruptions such as the April-2025 tariff shock.

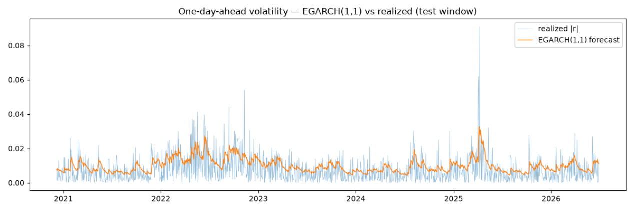

Fig. 3. Best model by QLIKE, EGARCH(1,1), against realized volatility over the test window.

ceilings may be somewhat higher than reported; with the LSTM already within statistical noise of the winner, tuning could plausibly close the remaining gap. (iv) Single seed per learned model. (v) The DM test between the two best models selected on the same test data carries a mild selection bias; the EGARCH-versus-GARCH comparison ($p < 0.0001$) is unaffected by this caveat.

6. CONCLUSION AND FUTURE WORK

Under a leak-proof walk-forward protocol on 18.5 years of S&P 500 data, a monthly re-estimated EGARCH(1,1) delivered the best one-day-ahead variance forecasts by QLIKE, statistically tied with GJR-GARCH, and both leverage models significantly outperformed plain GARCH(1,1) on a test window containing two asymmetric crashes. No learned model outranked the leverage GARCH family on the risk-relevant criterion, so the answer to the project's question remains a supported *no*, consistent with [7], but a narrower one than on shorter samples: the LSTM ranked third, ahead of plain GARCH, statistically indistinguishable from the winner, and the learned models took both point-accuracy metrics (LSTM the RMSE, XGBoost the MAE).

The main learning outcomes were methodological: constructing a genuinely leak-free walk-forward harness (target buffering, train-only scaling, end-of-train validation) proved harder and more consequential than model building, and the choice of loss function changed the ranking of models, a reminder that "better" is undefined until the metric is.

Future work: realized-variance targets from intraday data (the HAR-RV setting [8]), exogenous features such as the VIX term structure, multi-asset pooling to enlarge the effective training set, hybrid GARCH-plus-ML residual models, probabilistic forecasts under proper scoring rules, and robustness runs across assets (Nifty 50, gold, BTC) and across seeds.

7. ARTIFACTS AND DEMONSTRATIONS

- **Code repository:** `https://github.com/iitgF/T9`. Contains the modular `volforecast` Python package (configuration, data, metrics, features, baselines, XGBoost, LSTM, evaluation, process-parallel pipeline) and both notebooks.
- **LaTeX source (Overleaf):** `https://www.overleaf.com/project/6a48e18e7dee3049b3ee4209`
- **Video demonstration:** `ForecastingDailyEquityVolatility:CanMachineLearningBeatClassicalModels?`
- **Consolidated notebook:** `volatility_forecasting_consolidated.ipynb`, the full study in a single annotated notebook, every module inline with a detailed introduction, executed end-to-end with all results and figures reproduced in this report.
- **Starter/development notebook:** `volatility_forecasting_starter.ipynb`, importing from `volforecast`.

Attribution note: AI-assisted tooling (Claude Code) was used for code refactoring, optimization and drafting support. All modelling decisions, verification of results and the final text are the author's responsibility, and all reported numbers were reproduced end-to-end from the committed code.

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